1) a)

$3\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)$

1) b)

$5\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)$

1) c)

$2\left(\cos\left(-\frac{\pi}{3}\right)+\mathrm{i}\sin\left(-\frac{\pi}{3}\right)\right)=2\left(\cos\frac{2\pi}{3}+\mathrm{i}\sin\frac{2\pi}{3}\right)$

1) d)

$7\left(\cos\frac{18\pi}{5}+\mathrm{i}\sin\frac{18\pi}{5}\right)=7\left(\cos\left(-\frac{2\pi}{5}\right)+\mathrm{i}\sin\left(-\frac{2\pi}{5}\right)\right)$

1) e)

$-4\left(\cos\frac{3\pi}{4}+\mathrm{i}\sin\frac{3\pi}{4}\right)=4\left(\cos\left(-\frac{\pi}{4}\right)+\mathrm{i}\sin\left(-\frac{\pi}{4}\right)\right)$

1) f)

$-2\left(\cos\frac{5\pi}{4}+\mathrm{i}\sin\frac{5\pi}{4}\right)=2\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)$

2) a)

$2\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)=\sqrt{2}+\sqrt{2}\mathrm{i}$

2) b)

$3(\cos\pi+\mathrm{i}\sin\pi)=-3$

2) c)

$6\left(\cos\frac{7\pi}{3}+\mathrm{i}\sin\frac{7\pi}{3}\right)=3+3\sqrt{3}\mathrm{i}$

2) d)

$2(\cos(-\pi)+\mathrm{i}\sin(-\pi))=-2$

2) e)

$-4\left(\cos\frac{5\pi}{2}+\mathrm{i}\sin\frac{5\pi}{2}\right)=-4\mathrm{i}$

2) f)

$-\sqrt{2}\left(\cos\left(-\frac{4\pi}{3}\right)+\mathrm{i}\sin\left(-\frac{4\pi}{3}\right)\right)=\frac{\sqrt{2}}{2}-\frac{\sqrt{6}}{2}\mathrm{i}$

3) a)

$4\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}$

3) b)

$5\mathrm{e}^{-\frac{\pi}{4}\mathrm{i}}$

3) c)

$\sqrt{2}\mathrm{e}^{-\frac{4\pi}{5}\mathrm{i}}$

3) d)

$5\left(\cos\left(-\frac{3\pi}{2}\right)+\mathrm{i}\sin\left(-\frac{3\pi}{2}\right)\right)=5\left(\cos\left(\frac{\pi}{2}\right)+\mathrm{i}\sin\left(\frac{\pi}{2}\right)\right)=5\mathrm{e}^{\frac{\pi}{2}\mathrm{i}}$

3) e)

$4\left(\cos\frac{\pi}{6}-\mathrm{i}\sin\frac{\pi}{6}\right)=4\left(\cos\left(-\frac{\pi}{6}\right)+\mathrm{i}\sin\left(-\frac{\pi}{6}\right)\right)=4\mathrm{e}^{-\frac{\pi}{6}\mathrm{i}}$

3) f)

$\sqrt{3}\left(\cos\frac{6\pi}{7}-\mathrm{i}\sin\frac{6\pi}{7}\right)=\sqrt{3}\left(\cos\left(-\frac{6\pi}{7}\right)+\mathrm{i}\sin\left(-\frac{6\pi}{7}\right)\right)=\sqrt{3}\mathrm{e}^{-\frac{6\pi}{7}\mathrm{i}}$

3) g)

$\cos\left(-\frac{\pi}{5}\right)-\mathrm{i}\sin\left(-\frac{\pi}{5}\right)=\cos\left(\frac{\pi}{5}\right)+\mathrm{i}\sin\left(\frac{\pi}{5}\right)=\mathrm{e}^{\frac{\pi}{5}\mathrm{i}}$

3) h)

$8\left(\cos\left(-\frac{5\pi}{4}\right)-\mathrm{i}\sin\left(-\frac{5\pi}{4}\right)\right)=8\left(\cos\left(\frac{5\pi}{4}\right)+\mathrm{i}\sin\left(\frac{5\pi}{4}\right)\right)$

$=8\left(\cos\left(-\frac{3\pi}{4}\right)+\mathrm{i}\sin\left(-\frac{3\pi}{4}\right)\right)=8\mathrm{e}^{-\frac{3\pi}{4}\mathrm{i}}$

4) a)

$|z| = \sqrt{(2)^2 + (2)^2} = 2\sqrt{2}$ and $\arg z = \arctan\left(\frac{\sqrt{2}}{\sqrt{2}}\right) = \frac{\pi}{4}$

$\therefore z = 2\sqrt{2}\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}$

4) b)

$|z| = \sqrt{4^2 + \left(-4\sqrt{3}\right)^2} = 8$ and $\arg z = -\arctan\left(\frac{4\sqrt{3}}{4}\right) = -\frac{\pi}{3}$

$\therefore z = 8\mathrm{e}^{-\frac{\pi}{3}\mathrm{i}}$

4) c)

$|z| = \sqrt{(-2)^2 + \left(\sqrt{2}\right)^2} = \sqrt{6}$ and $\arg z = \pi - \arctan\left(\frac{\sqrt{2}}{2}\right) = 2.53$

$\therefore z = \sqrt{6}\mathrm{e}^{2.53\mathrm{i}}$

4) d)

$|z| = \sqrt{(-5)^2 + (-12)^2} = 13$ and $\arg z = -\pi + \arctan\left(\frac{12}{5}\right) = -1.97$

$\therefore z = 13\mathrm{e}^{-1.97\mathrm{i}}$

4) e)

$|z| = 7$ and $\arg z = \pi$

$\therefore z = 7\mathrm{e}^{\pi\mathrm{i}}$

4) f)

$|z| = 6$ and $\arg z = \frac{\pi}{2}$

$\therefore z = 6\mathrm{e}^{\frac{\pi}{2}\mathrm{i}}$

4) g)

$|z| = 8$ and $\arg z = 0$

$\therefore z = 8\mathrm{e}^{0\mathrm{i}}$

4) h)

$|z| = \sqrt{3^2 + 1^2} = \sqrt{10}$ and $\arg z = \arctan\left(\frac{1}{3}\right) = 0.322$

$\therefore z = \sqrt{10}\mathrm{e}^{0.322\mathrm{i}}$

4) i)

$|z| = \sqrt{1^2 + \left(-\sqrt{5}\right)^2} = \sqrt{6}$ and $\arg z = -\arctan\left(\frac{\sqrt{5}}{1}\right) = -1.15$

$\therefore z = \sqrt{6}\mathrm{e}^{-1.15\mathrm{i}}$

5) a)

$z^* = 2\mathrm{e}^{-\frac{\pi}{5}\mathrm{i}}$

5) b)

$z = 8\mathrm{e}^{-\frac{\pi}{2}\mathrm{i}}$

So $z^* = 8\mathrm{e}^{\frac{\pi}{2}\mathrm{i}}$

5) c)

$z = 3\mathrm{e}^{-\frac{6\pi}{7}\mathrm{i}}$

So $z^* = 3\mathrm{e}^{\frac{6\pi}{7}\mathrm{i}}$

5) d)

$z = -6\mathrm{e}^{\frac{2\pi}{7}\mathrm{i}} = 6\mathrm{e}^{-\frac{5\pi}{7}\mathrm{i}}$

So $z^* = 6\mathrm{e}^{\frac{5\pi}{7}\mathrm{i}}$

6) a)

$\mathrm{e}^{\frac{\pi}{4}\mathrm{i}+\frac{\pi}{2}\mathrm{i}} = \mathrm{e}^{\frac{3\pi}{4}\mathrm{i}} = \cos\frac{3\pi}{4} + \mathrm{i}\sin\frac{3\pi}{4} = -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\mathrm{i}$

6) b)

$2\sqrt{3}\mathrm{e}^{\frac{\pi}{3}\mathrm{i}+\frac{\pi}{5}\mathrm{i}} = 2\sqrt{3}\mathrm{e}^{\frac{8\pi}{15}\mathrm{i}} = 2\sqrt{3}\left(\cos\frac{8\pi}{15} + \mathrm{i}\sin\frac{8\pi}{15}\right) = -0.362 + 3.45\mathrm{i}$

6) c)

$-2e^{2i+5i} = -2e^{7i} = -2(\cos 7 + i \sin 7) = -1.51 - 1.31i$

6) d)

$e^{\frac{\pi}{4}i-\frac{\pi}{2}i+\frac{\pi}{3}i} = e^{\frac{\pi}{12}i} = \cos\frac{\pi}{12} + i\sin\frac{\pi}{12} = \frac{\sqrt{6}+\sqrt{2}}{4} + \left(\frac{\sqrt{6}-\sqrt{2}}{4}\right)i$

6) e)

$2\sqrt{2}e^{-\frac{\pi}{6}i+\frac{\pi}{3}i+\frac{5\pi}{4}i} = 2\sqrt{2}e^{\frac{17\pi}{12}i} = 2\sqrt{2}\left(\cos\frac{17\pi}{12} + i\sin\frac{17\pi}{12}\right) = 1 - \sqrt{3} - (1+\sqrt{3})i$

7) a)

$e^{\frac{\pi}{4}i-\frac{\pi}{2}i} = e^{-\frac{\pi}{4}i} = \cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i$

7) b)

$\frac{1}{3}e^{\frac{\pi}{3}i-\frac{\pi}{4}i} = \frac{1}{3}e^{\frac{\pi}{12}i} = \frac{1}{3}\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right) = \frac{\sqrt{6}+\sqrt{2}}{12} + \left(\frac{\sqrt{6}-\sqrt{2}}{12}\right)i$

7) c)

$\frac{1}{5}e^{3i-7i} = \frac{1}{5}e^{-4i} = \frac{1}{5}(\cos(-4) + i\sin(-4)) = -0.131 + 0.151i$

7) d)

$4e^{\frac{\pi}{3}i+\frac{\pi}{4}i-\frac{\pi}{2}i} = 4e^{\frac{\pi}{12}i} = 4\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right) = \sqrt{6} + \sqrt{2} + (\sqrt{6} - \sqrt{2})i$

7) e)

$\frac{\sqrt{2}}{3}e^{-\frac{\pi}{4}i-\frac{\pi}{5}i+\frac{4\pi}{7}i} = \frac{\sqrt{2}}{3}e^{\frac{17\pi}{140}i} = \frac{\sqrt{2}}{3}\left(\cos\frac{17\pi}{140} + i\sin\frac{17\pi}{140}\right) = 0.438 + 0.176i$

8)

$e^{3\theta i} \times e^{2\theta i} = e^{5\theta i}$

9) a)

$e^{\frac{3\pi}{7}i} \times e^{\frac{4\pi}{7}i} = e^{\pi i}$

9) b)

$$e^{\frac{\pi}{4}i} \times e^{-\frac{\pi}{3}i} = e^{-\frac{\pi}{12}i}$$

9) c)

$$3e^{-\frac{\pi}{4}i} \times e^{\frac{3\pi}{8}i} = 3e^{\frac{\pi}{8}i}$$

9) d)

$$4\left(\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right)\left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right) = 4e^{-\frac{\pi}{3}i} \times e^{-\frac{\pi}{4}i} = 4e^{-\frac{7\pi}{12}i}$$

10)

$$\frac{e^{6\theta i}}{e^{4\theta i}} = e^{2\theta i}$$

11) a)

$$\frac{e^{\frac{2\pi}{5}i}}{e^{\frac{4\pi}{5}i}} = e^{-\frac{2\pi}{5}i}$$

11) b)

$$\frac{e^{\frac{\pi}{4}i}}{e^{-\frac{2\pi}{3}i}} = e^{\frac{11\pi}{12}i}$$

11) c)

$$\frac{\cos\frac{\pi}{5}+i\sin\frac{\pi}{5}}{\cos\frac{\pi}{3}-i\sin\frac{\pi}{3}} = \frac{\cos\frac{\pi}{5}+i\sin\frac{\pi}{5}}{\cos\left(-\frac{\pi}{3}\right)+i\sin\left(-\frac{\pi}{3}\right)} = \frac{e^{\frac{\pi}{5}i}}{e^{-\frac{\pi}{3}i}} = e^{\frac{8\pi}{15}i}$$

12) a)

$$|z| = \sqrt{2^2 + (-2)^2} = 2\sqrt{2}$$

$$\arg z = -\arctan\left(\frac{2}{2}\right) = -\frac{\pi}{4}$$

$$\therefore z = 2\sqrt{2}e^{-\frac{\pi}{4}i}$$

12) b)

$|w| = \sqrt{1^2 + \left(\sqrt{3}\right)^2} = 2$

$\arg w = \arctan\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3}$

$\therefore w = 2e^{\frac{\pi}{3}i}$

12) c)

$z^2 + w^2 = \left(2\sqrt{2}e^{-\frac{\pi}{4}i}\right)^2 + \left(2e^{\frac{\pi}{3}i}\right)^2 = 8e^{-\frac{\pi}{2}i} + 4e^{\frac{2\pi}{3}i}$

$= 8\left(\cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)\right) + 4\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$

So that

$a = 8\cos\left(-\frac{\pi}{2}\right) + 4\cos\frac{2\pi}{3} = -2$

and $b = 8\sin\left(-\frac{\pi}{2}\right) + 4\sin\frac{2\pi}{3} = -8 + 2\sqrt{3}$

Which gives:

$\frac{b}{a} = \frac{-8+2\sqrt{3}}{-2} = 4 - \sqrt{3}$ as req.

13) a)

$3\sqrt{2}$

13) b)

$\frac{3}{\sqrt{2}}$ or $\frac{3\sqrt{2}}{2}$

13) c)

$\frac{\pi}{4} + \frac{\pi}{3} = \frac{7\pi}{12}$

13) d)

$\frac{\pi}{4} - \frac{\pi}{3} = -\frac{\pi}{12}$

14) a)

$3\sqrt{3}$

14) b)

$\frac{\sqrt{3}}{3}$

14) c)

$-\frac{2\pi}{3}+\frac{\pi}{5}=-\frac{7\pi}{15}$

14) d)

$-\frac{2\pi}{3}-\frac{\pi}{5}=-\frac{13\pi}{15}$

15) a)

$z_1z_2=r_1\mathrm{e}^{\theta_1\mathrm{i}}\times r_2\mathrm{e}^{\theta_2\mathrm{i}}=r_1r_2\mathrm{e}^{(\theta_1+\theta_2)\mathrm{i}}$

So $|z_1z_2|=r_1r_2=|z_1||z_2|$ as req.

15) b)

$\frac{z_1}{z_2}=\frac{r_1\mathrm{e}^{\theta_1\mathrm{i}}}{r_2\mathrm{e}^{\theta_2\mathrm{i}}}=\frac{r_1}{r_2}\mathrm{e}^{(\theta_1-\theta_2)\mathrm{i}}$

So $\left|\frac{z_1}{z_2}\right|=\frac{r_1}{r_2}=\frac{|z_1|}{|z_2|}$ as req.

15) c)

$z_1z_2=r_1\mathrm{e}^{\theta_1\mathrm{i}}\times r_2\mathrm{e}^{\theta_2\mathrm{i}}=r_1r_2\mathrm{e}^{(\theta_1+\theta_2)\mathrm{i}}$

So $\arg(z_1z_2)=\theta_1+\theta_2=\arg(z_1)+\arg(z_2)$ as req.

15) d)

$\frac{z_1}{z_2}=\frac{r_1\mathrm{e}^{\theta_1\mathrm{i}}}{r_2\mathrm{e}^{\theta_2\mathrm{i}}}=\frac{r_1}{r_2}\mathrm{e}^{(\theta_1-\theta_2)\mathrm{i}}$

So $\arg\left(\frac{z_1}{z_2}\right)=\theta_1-\theta_2=\arg(z_1)-\arg(z_2)$ as req.

16) a)

$|z| = \sqrt{(-6)^2 + \left(6\sqrt{3}\right)^2} = 12$

$\arg z = \pi - \arctan\left(\frac{6\sqrt{3}}{6}\right) = \frac{2\pi}{3}$

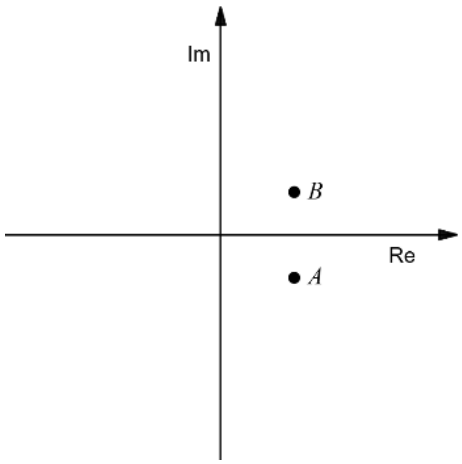
So $z = 12\mathrm{e}^{\frac{2\pi}{3}\mathrm{i}}$ and $w = 4\mathrm{e}^{-\frac{5\pi}{12}\mathrm{i}}$

$zw = 12\mathrm{e}^{\frac{2\pi}{3}\mathrm{i}} \times 4\mathrm{e}^{-\frac{5\pi}{12}\mathrm{i}} = 48\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}$

16) b)

$\frac{z}{w} = \frac{12\mathrm{e}^{\frac{2\pi}{3}\mathrm{i}}}{4\mathrm{e}^{-\frac{5\pi}{12}\mathrm{i}}} = 3\mathrm{e}^{\frac{13\pi}{12}\mathrm{i}} = 3\mathrm{e}^{-\frac{11\pi}{12}\mathrm{i}}$

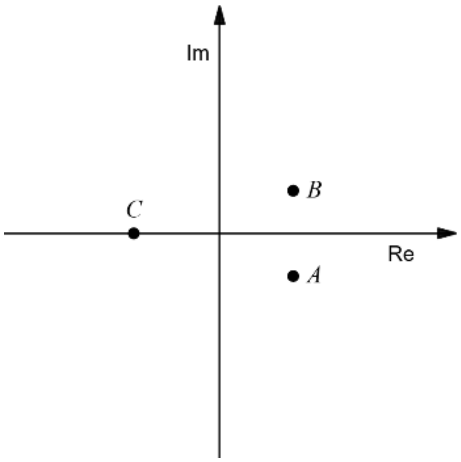
17) a)



17) b)

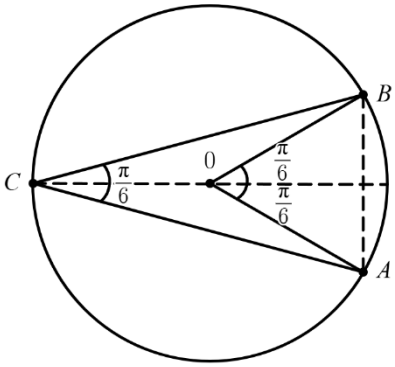
A is given by $4\left(\cos\left(-\frac{\pi}{6}\right)+\mathrm{i}\sin\left(-\frac{\pi}{6}\right)\right)=2\sqrt{3}-2\mathrm{i}$

B is its complex conjugate: $2\sqrt{3}+2\mathrm{i}$

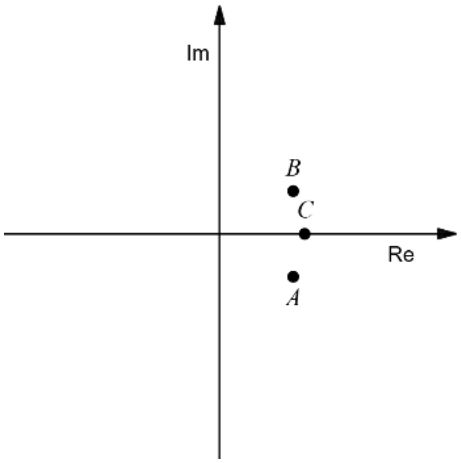


Since all of the points have a modulus of 4, they all lie on a circle:

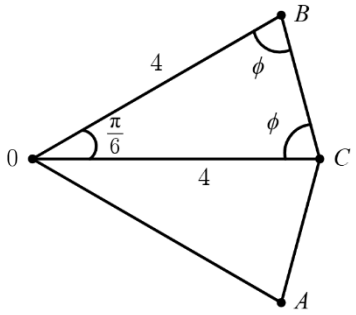
If C is -4 then



And so $\angle ACB=2\times\frac{\pi}{12}=\frac{\pi}{6}$



If C is 4 then



And so $\angle ACB = \frac{1}{2}\left(\pi - \frac{\pi}{6}\right) \times 2 = \frac{5\pi}{6}$

So the possible angles are $\frac{\pi}{6}$ or $\frac{5\pi}{6}$

18) a)

$\frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right) = \frac{1}{2}e^{i\theta} + \frac{1}{2}e^{-i\theta}$

$= \frac{1}{2}(\cos\theta + i\sin\theta) + \frac{1}{2}(\cos\theta - i\sin\theta)$

$= \frac{1}{2}\cos\theta + \frac{1}{2}i\sin\theta + \frac{1}{2}\cos\theta - \frac{1}{2}i\sin\theta$

$= \cos\theta$ as req.

18) b)

$\frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right) \times \frac{i}{i} = -\frac{i}{2}\left(e^{i\theta} - e^{-i\theta}\right)$

$= -\frac{i}{2}e^{i\theta} + \frac{i}{2}e^{-i\theta}$

$= -\frac{i}{2}(\cos\theta + i\sin\theta) + \frac{i}{2}(\cos\theta - i\sin\theta)$

$= -\frac{i}{2}\cos\theta - \frac{i^2}{2}\sin\theta + \frac{i}{2}\cos\theta - \frac{i^2}{2}\sin\theta$

$= \frac{1}{2}\sin\theta + \frac{1}{2}\sin\theta$

$= \sin\theta$ as req.

Alternatively:

$e^{i\theta} = \cos\theta + i\sin\theta$ (1)

$e^{-i\theta} = \cos(-\theta) + i\sin(-\theta)$

$= \cos\theta - i\sin\theta$ (2)

(1) + (2) $e^{i\theta} + e^{-i\theta} = 2\cos\theta$

$\Rightarrow \cos\theta = \frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right)$

(1) - (2) $e^{i\theta} - e^{-i\theta} = 2i\sin\theta$

$\Rightarrow \sin\theta = \frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right)$

19) a)

$e^{i\theta} = \cos\theta + i\sin\theta$ and $e^{-i\theta} = \cos\theta - i\sin\theta$

Multiply the LHS:

$e^{i\theta} \times e^{-i\theta} = 1$

Multiply the RHS:

$(\cos\theta + i\sin\theta)(\cos\theta - i\sin\theta)$

$= \cos^2\theta + i\sin\theta\cos\theta - i\sin\theta\cos\theta + \sin^2\theta$

$= \cos^2\theta + \sin^2\theta$

$\therefore \sin^2\theta + \cos^2\theta = 1$ as req.

19) b)

Since $e^{i\theta} = \cos\theta + i\sin\theta$ and $e^{-i\theta} = \cos\theta - i\sin\theta$

Then $e^{i\theta} + e^{-i\theta} = 2\cos\theta$ so $\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$

and $e^{i\theta} - e^{-i\theta} = 2i\sin\theta$ so $\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$

So that:

$2\sin\theta\cos\theta = 2 \times \frac{1}{2i}(e^{i\theta} - e^{-i\theta}) \times \frac{1}{2}(e^{i\theta} + e^{-i\theta})$

$= \frac{1}{2i}(e^{2i\theta} - e^{-2i\theta})$

$= \sin 2\theta$ as req.

19) c)

Since $e^{i\theta} = \cos\theta + i\sin\theta$ and $e^{-i\theta} = \cos\theta - i\sin\theta$

Then $e^{i\theta} + e^{-i\theta} = 2\cos\theta$ so $\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$

and $e^{i\theta} - e^{-i\theta} = 2i\sin\theta$ so $\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$

So that:

$\cos^2\theta - \sin^2\theta = \frac{1}{4}(e^{i\theta} + e^{-i\theta})^2 + \frac{1}{4}(e^{i\theta} - e^{-i\theta})^2$

$= \frac{1}{4}(e^{2i\theta} + 2 + e^{-2i\theta}) + \frac{1}{4}(e^{2i\theta} - 2 + e^{-2i\theta})$

$= \frac{1}{2}(e^{2i\theta} + e^{-2i\theta})$

$= \cos 2\theta$ as req.

20)

$(\cos\theta + i\sin\theta)^n = (e^{i\theta})^n = e^{in\theta} = \cos n\theta + i\sin n\theta$

21)

$e^{z} = e^{a+bi} = e^{a}e^{bi} = e^{a}(\cos b + i \sin b)$

$e^{z^*} = e^{a-bi} = e^{a}e^{-bi} = e^{a}(\cos b - i \sin b)$

$(e^{z})^* = e^{a}(\cos b - i \sin b)$

$\therefore e^{z^*} = (e^{z})^*$ as req.

Alternatively:

$e^{z^*} = e^{a-bi} = e^{a}e^{-bi}$

$= e^{a}(\cos(-b) + i \sin(-b))$

$= e^{a}(\cos b - i \sin b)$

$= e^{a}\left(e^{bi}\right)^* = \left(e^{a+bi}\right)^* = (e^{z})^*$

22)

$e^{i\theta} = \cos\theta + i \sin\theta$ and $e^{-i\theta} = \cos\theta - i \sin\theta$

Factor out $e^{i\theta}$:

$e^{2i\theta} - 1 = e^{i\theta}\left(e^{i\theta} - e^{-i\theta}\right)$

Since $e^{i\theta} - e^{-i\theta} = 2i \sin\theta$:

$e^{2i\theta} - 1 = e^{i\theta}(2i \sin\theta)$

$= 2ie^{i\theta} \sin\theta$ as req.

23)

$\left(1 - \frac{1}{4}e^{i\theta}\right)\left(1 - \frac{1}{4}e^{-i\theta}\right)$

$= 1 - \frac{1}{4}e^{i\theta} - \frac{1}{4}e^{-i\theta} + \frac{1}{16}$

$= \frac{17}{16} - \frac{1}{2} \times \frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right)$

$= \frac{17}{16} - \frac{1}{2}\cos\theta$

24)

$\left(1 - ae^{i\theta}\right)\left(1 - ae^{-i\theta}\right)$

$= 1 - ae^{i\theta} - ae^{-i\theta} + a^2$

$= 1 + a^2 - 2a \times \frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right)$

$= 1 + a^2 - 2a\cos\theta$

25)

$z + w = 2e^{\frac{\pi}{6}i} + 2e^{\frac{\pi}{3}i} = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right) + 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

$= 1 + \sqrt{3} + (1 + \sqrt{3})i$

$|z| = \sqrt{(1+\sqrt{3})^2 + (1+\sqrt{3})^2} = \sqrt{2(1+\sqrt{3})^2} = \sqrt{2}(1+\sqrt{3})$

$\arg z = \arctan\left(\frac{1+\sqrt{3}}{1+\sqrt{3}}\right) = \frac{\pi}{4}$

$\therefore z = \sqrt{2}(1+\sqrt{3})e^{\frac{\pi}{4}i} = (\sqrt{2}+\sqrt{6})e^{\frac{\pi}{4}i}$ as req.